

Installation Guide

LogiSource Integrated System

Complete step-by-step guide to install LogiSource on your local PC or deploy it to a production VPS / hosting. Covers Windows, macOS, Linux, plus optional Docker and static frontend hosting on Netlify / Vercel / S3.

Backend	FastAPI 0.11x · Python 3.11+ · Motor (async MongoDB driver)
Frontend	React 19 · TailwindCSS · Shadcn UI · Craco
Database	MongoDB 5.0+
Auth	JWT (bcrypt) · AES-256-GCM vault
Docs	PDF via ReportLab · Excel via openpyxl

LogiSource Digital

Generated 23 July 2026 · 09:22 UTC

Contents

Part A	Local PC installation
1	Pre-requirements (all platforms)
2	Install on Windows
3	Install on macOS
4	Install on Linux (Ubuntu / Debian)
5	First run & login
Part B	Hosting / VPS deployment
6	Server requirements
7	Domain, DNS & HTTPS
8	Full VPS setup (Ubuntu step-by-step)
9	nginx reverse proxy configuration
10	systemd service files
11	Docker deployment (optional)
12	Static frontend on Netlify / Vercel / S3
Part C	Operations
13	Environment variables reference
14	Backups & disaster recovery
15	Troubleshooting

Part A · Local PC installation

This part gets LogiSource running on a developer's laptop or a single office PC. You'll extract the ZIP you downloaded from Website Settings → Download ZIP, install three dependencies (Python, Node.js and MongoDB), then start the backend and frontend as two separate processes.

1 · Pre-requirements (all platforms)

Tool	Version	Purpose
Python	>= 3.11	Runs the FastAPI backend
pip	bundled with Python	Installs backend Python packages
Node.js	>= 18.x (LTS)	Runs the React frontend build
Yarn	>= 1.22	Package manager for the frontend (npm is NOT supported)
MongoDB	>= 5.0	Database (Community edition is fine)
mongodump/mongorestore	any recent	For backup / restore of BSON dumps
git	any recent	Optional — only if you plan to push code to a repo

Recommended hardware

2 CPU cores · 4 GB RAM · 5 GB free disk. LogiSource itself is lightweight — most memory is spent on MongoDB (~500 MB idle) and the React dev server (~500 MB).

■ Do **not** use `npm install` for the frontend. This project uses Yarn (see `yarn.lock`).

2 · Install on Windows

2.1 Install Python 3.11+

Download the installer from python.org/downloads. In the installer, tick **“Add python.exe to PATH”** before clicking Install. Verify:

```
python --version
pip --version
```

2.2 Install Node.js LTS + Yarn

Download the LTS installer from nodejs.org. After installation, open a new PowerShell as Administrator and enable Yarn via Corepack:

```
corepack enable
corepack prepare yarn@stable --activate
yarn --version
```

2.3 Install MongoDB Community

Download MongoDB Community Server from mongodb.com/try/download/community. During install pick **“Complete”** and keep **“Install MongoDB as a Service”** ticked. MongoDB will run as a Windows Service on port 27017. Verify with MongoDB Compass or:

```
"C:\Program Files\MongoDB\Server\7.0\bin\mongosh.exe" --eval "db.runCommand({ping:1})"
```

2.4 Extract the ZIP and set up the backend

```
cd C:\LogiSource
Expand-Archive .\logisource-full-*.zip . # or right-click → Extract All

cd backend
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt

# Restore the database (BSON dump)
mongorestore --uri "mongodb://localhost:27017" --db test_database ..\database\test_database

# Start the backend
uvicorn server:app --host 0.0.0.0 --port 8001
```

2.5 Start the frontend

Open a second PowerShell window:

```
cd C:\LogiSource\frontend
yarn install # once, first time only
yarn start # opens http://localhost:3000
```

3 · Install on macOS

The easiest path is Homebrew.

3.1 Install Homebrew (if you don't have it)

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

3.2 Install runtimes

```
brew install python@3.11 node yarn
brew tap mongodb/brew
brew install mongodb-community@7.0 mongodb-database-tools
```

```
# Start MongoDB as a background service
brew services start mongodb-community@7.0
```

3.3 Backend + frontend

```
cd ~/Downloads && unzip logisource-full-*.zip -d ~/LogiSource
cd ~/LogiSource/backend
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
mongorestore --uri "mongodb://localhost:27017" --db test_database ../database/test_database
uvicorn server:app --host 0.0.0.0 --port 8001 &

cd ../frontend
yarn install && yarn start
```

4 · Install on Linux (Ubuntu 22.04 / Debian 12)

4.1 Install runtimes

```
sudo apt update && sudo apt install -y python3-venv python3-pip curl git
```

```
# Node.js 20 LTS + Yarn via Corepack
```

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
sudo corepack enable
```

```
# MongoDB 7 (Ubuntu 22.04)
```

```
curl -fsSL https://pgp.mongodb.com/server-7.0.asc | \
    sudo gpg -o /usr/share/keyrings/mongodb-server-7.0.gpg --dearmor
echo "deb [ signed-by=/usr/share/keyrings/mongodb-server-7.0.gpg ] https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org-7.0 bin" | \
    sudo tee /etc/apt/sources.list.d/mongodb-org-7.0.list
sudo apt update && sudo apt install -y mongodb-org
sudo systemctl enable --now mongod
```

4.2 Extract and run

```
unzip logisource-full-*.zip -d ~/LogiSource
cd ~/LogiSource/backend
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
mongorestore --uri "mongodb://localhost:27017" --db test_database ../database/test_database
uvicorn server:app --host 0.0.0.0 --port 8001 &

cd ../frontend
yarn install && yarn start
```

5 · First run & login

With backend on :8001 and frontend on :3000, open <http://localhost:3000> in your browser. Log in with the seed credentials:

Role	Email	Password
Administrator	admin@logisource.com	Admin@12345

Staff	staff@logisource.com	Staff@12345
-------	----------------------	-------------

■ Immediately change both passwords from the top-right menu → **Change Password**, and open Website Settings to configure your company name, tax %, and prefixes.

Part B · Hosting / VPS deployment

This part covers a real production deployment on a Linux VPS with a domain, HTTPS via Let's Encrypt, and process management via systemd.

6 · Server requirements

Component	Minimum	Recommended
CPU	1 vCPU	2 vCPU
RAM	2 GB	4 GB
Disk	10 GB SSD	20 GB SSD
OS	Ubuntu 22.04 LTS	Ubuntu 22.04 LTS
Network	Public IPv4	Public IPv4 + IPv6
Ports open	80, 443 (public); 8001 & 27017 loopback only	

Popular VPS providers: DigitalOcean (\$6/mo droplet), Hetzner (€4/mo CX22), Vultr, Linode, or any on-prem server.

7 · Domain, DNS & HTTPS

Point one subdomain (e.g. **logisource.yourdomain.com**) to the server's public IP with an A record. Optionally use a second subdomain for the backend API (e.g. **api.logisource.yourdomain.com**) — but a single subdomain with an nginx reverse-proxy for both `/` (frontend) and `/api/` (backend) is simpler.

HTTPS is mandatory in production because the JWT auth cookie has Secure flag on. Use certbot for a free Let's Encrypt certificate (installed in section 8).

8 · Full VPS setup (Ubuntu 22.04 step-by-step)

SSH into the fresh server as root, then:

8.1 Base hardening & users

```
adduser deploy
usermod -aG sudo deploy
rsync --archive --chown=deploy:deploy ~/.ssh /home/deploy
# Now log in as deploy@server and continue

sudo apt update && sudo apt upgrade -y
sudo apt install -y ufw
sudo ufw allow OpenSSH && sudo ufw allow 80 && sudo ufw allow 443
sudo ufw --force enable
```

8.2 Install runtimes (same as Section 4.1)

Repeat the Ubuntu install script from section 4.1 (Python, Node, Yarn, MongoDB).

8.3 Upload & extract LogiSource

```
sudo mkdir -p /opt/logisource && sudo chown deploy:deploy /opt/logisource
cd /opt/logisource
# Upload logisource-full-*.zip via scp/sftp, then:
unzip logisource-full-*.zip
ls # should show backend/ frontend/ database/ memory/ README.md
```

8.4 Rotate secrets (mandatory!)

```
cd /opt/logisource/backend

# Generate strong values
python3 -c "import secrets; print('JWT_SECRET='+secrets.token_hex(32))"
python3 -c "import secrets; print('VAULT_MASTER_KEY='+secrets.token_hex(32))"

# Edit .env — REPLACE JWT_SECRET, VAULT_MASTER_KEY, ADMIN_PASSWORD, STAFF_PASSWORD
```

■ If you rotate VAULT_MASTER_KEY on an existing database, all encrypted vault passwords become unreadable. Rotate BEFORE users add credentials, or plan a decrypt-re-encrypt migration.

8.5 Restore database

```
cd /opt/logisource/backend
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt

mongorestore --uri "mongodb://localhost:27017" --db test_database \
  ../database/test_database
```

8.6 Build the frontend for production

```
cd /opt/logisource/frontend

# Point the compiled bundle at your public backend URL
echo "REACT_APP_BACKEND_URL=https://logisource.yourdomain.com" > .env

yarn install
yarn build # output → /opt/logisource/frontend/build/
```


9 · nginx reverse proxy

```
sudo apt install -y nginx certbot python3-certbot-nginx
```

Create /etc/nginx/sites-available/logisource:

```
server {
    listen 80;
    server_name logisource.yourdomain.com;

    # ---- Frontend (static build) ----
    root /opt/logisource/frontend/build;
    index index.html;
    location / {
        try_files $uri /index.html;
    }

    # ---- Backend API ----
    location /api/ {
        proxy_pass http://127.0.0.1:8001;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_read_timeout 300s;
        client_max_body_size 50m;
    }
}

sudo ln -s /etc/nginx/sites-available/logisource /etc/nginx/sites-enabled/
sudo nginx -t && sudo systemctl reload nginx
```

Automatic HTTPS via Let's Encrypt

```
sudo certbot --nginx -d logisource.yourdomain.com --redirect --agree-tos -m you@yourdomain.com
```

certbot will rewrite the nginx config to listen on 443 with a valid TLS certificate and auto-renew via a systemd timer. Test <https://logisource.yourdomain.com> in your browser.

10 · systemd service for the backend

Create /etc/systemd/system/logisource-backend.service:

```
[Unit]
Description=LogiSource FastAPI backend
After=network.target mongod.service
Requires=mongod.service

[Service]
Type=simple
User=deploy
Group=deploy
WorkingDirectory=/opt/logisource/backend
Environment="PATH=/opt/logisource/backend/.venv/bin"
EnvironmentFile=/opt/logisource/backend/.env
ExecStart=/opt/logisource/backend/.venv/bin/uvicorn server:app \
    --host 127.0.0.1 --port 8001 --workers 2 --proxy-headers
Restart=on-failure
RestartSec=3

[Install]
WantedBy=multi-user.target
sudo systemctl daemon-reload
sudo systemctl enable --now logisource-backend
sudo systemctl status logisource-backend
journalctl -u logisource-backend -f          # live logs
```

11 · Docker deployment (optional)

If you prefer containers, drop this docker-compose.yml into the extracted folder next to backend/ and frontend/:

```
services:
  mongo:
    image: mongo:7
    restart: unless-stopped
    volumes:
      - mongo-data:/data/db

  backend:
    image: python:3.11-slim
    working_dir: /app
    volumes:
      - ./backend:/app
    command: >
      sh -c "pip install -r requirements.txt &&
        uvicorn server:app --host 0.0.0.0 --port 8001 --workers 2"
    env_file: ./backend/.env
    environment:
      MONGO_URL: mongodb://mongo:27017
    depends_on: [mongo]
    ports: ["8001:8001"]

  frontend:
    image: node:20-alpine
    working_dir: /app
    volumes:
      - ./frontend:/app
    command: sh -c "corepack enable && yarn install && yarn start"
    environment:
      REACT_APP_BACKEND_URL: http://localhost:8001
    ports: ["3000:3000"]

volumes:
  mongo-data:

docker compose up -d
docker compose logs -f backend
```

12 · Static frontend on Netlify / Vercel / S3

The compiled frontend/build/ folder is a plain static site. You can deploy the backend wherever (Render, Railway, self-hosted VPS) and host the frontend for free on a CDN.

Netlify

```
# Once, on your machine
npm i -g netlify-cli # (netlify's own tool; still uses your yarn build)
cd frontend && yarn build
netlify deploy --prod --dir=build
```

Vercel

```
npm i -g vercel
cd frontend && vercel --prod
# In the Vercel dashboard, set env var: REACT_APP_BACKEND_URL
```

AWS S3 + CloudFront

```
aws s3 sync frontend/build/ s3://your-bucket/ --delete
# If you use CloudFront, don't forget to update the distribution
```

■ Whichever host you pick, remember to update **CORS_ORIGINS** in the backend .env to include the frontend's URL, or CORS-preflight requests will fail.

Part C · Operations

13 · Environment variables reference

Variable	Where	Purpose
MONGO_URL	backend/.env	mongodb://user:pass@host:27017
DB_NAME	backend/.env	Database name (default test_database — rename in prod!)
JWT_SECRET	backend/.env	Signs auth tokens. 64 hex chars. ROTATE for prod.
JWT_ALGORITHM	backend/.env	HS256 (do not change)
JWT_ACCESS_MINUTES	backend/.env	Token lifetime (default 720 = 12 hrs)
VAULT_MASTER_KEY	backend/.env	AES-256 key for the credential vault. ROTATE ONCE at setup.
ADMIN_EMAIL / _PASSWORD	backend/.env	Seeded on first start-up if user doesn't exist
STAFF_EMAIL / _PASSWORD	backend/.env	Same, for the seeded staff user
CORS_ORIGINS	backend/.env	Comma-separated list, e.g. https://logisource.yourdomain.com
COMPANY_NAME / _TAGLINE / _LOGO_URL	backend/.env	Default settings, editable in-app
REACT_APP_BACKEND_URL	frontend/.env	Baked into the build. Set BEFORE running yarn build.

14 · Backups & disaster recovery

Two built-in backup paths — both accessible from Website Settings → Backup & Restore:

- **Backup** — one-click JSON dump of every collection (small, portable, but not indexed).
- **Download ZIP** — full source + BSON MongoDB dump. Use for archival + off-server storage.

For scheduled off-server backups on a VPS, use this cron entry:

```
# /etc/cron.d/logisource-backup
0 3 * * * deploy mongodump --uri "mongodb://localhost:27017" --db test_database \
  --gzip --archive=/var/backups/logisource-$(date +%Y%m%d).gz
```

To restore any dump on a fresh install:

```
mongorestore --uri "mongodb://localhost:27017" --gzip \
  --archive=/var/backups/logisource-20260715.gz --nsInclude "test_database.*"
```

15 · Troubleshooting

Symptom	Cause / Fix
Login page loads but login POST returns 500	backend not running or MONGO_URL unreachable. Check <code>journalctl -u logisource-backend -f</code>
Login returns 401 with correct password	JWT_SECRET was rotated after the token was issued. Log out (or clear localStorage) and log in again.
Vault reveal shows blank string	VAULT_MASTER_KEY was changed after the record was created. Restore original key or re-create record.
CORS error in browser console	backend .env → CORS_ORIGINS must include the frontend URL (protocol + host, no trailing slash)
yarn install fails with ENOENT node-gyp	You are on npm. Delete node_modules + package-lock.json, then use <code>yarn install</code> .
nginx: 502 Bad Gateway on /api	Backend process died. <code>sudo systemctl restart logisource-backend</code> .

Slow PDF generation on VPS	Give the backend at least 512 MB RAM. Reduce unicorn --workers to 1 on tiny droplets.
Cookie not set / user logs out immediately	Frontend served over HTTP but backend cookie has Secure=true. Enforce HTTPS via certbot.

Need more help? Open the LogiSource web app → sidebar → **Activity Logs** to see every backend action with timestamp and user, or open your VPS logs via `journalctl -u logisource-backend`.